

FPGA2 Week10 Team3

2021041091_한규철, 2021041089_최태민

비동기식reset + 8진 segment counter + Beep 영상 : <https://youtu.be/DarWXztJbiU>

비동기식reset + 5진 segment counter + Beep 영상 : <https://youtu.be/IE1U33jugLM>

실습1 + 과제LOAD 기능추가: <https://youtu.be/h2nBaQFZ4as?si=Cy75qPw3K3OvoQFO>

실습2 + 과제LOAD 기능추가: <https://youtu.be/EpnJecLubIQ?si=l4l2YmURgBfpX1Tc>

동작영상 유튜브 주소

비동기식reset + 8진 segment counter + Beep 영상 : <https://youtu.be/DarWXztJbiU>

비동기식reset + 5진 segment counter + Beep 영상 : <https://youtu.be/IE1U33jugLM>

실습1 + 과제LOAD 기능추가: <https://youtu.be/h2nBaQFZ4as?si=Cy75qPw3K3OvoQFO>

실습2 + 과제LOAD 기능추가: <https://youtu.be/EpnJecLublQ?si=l4l2YmURgBfpX1Tc>

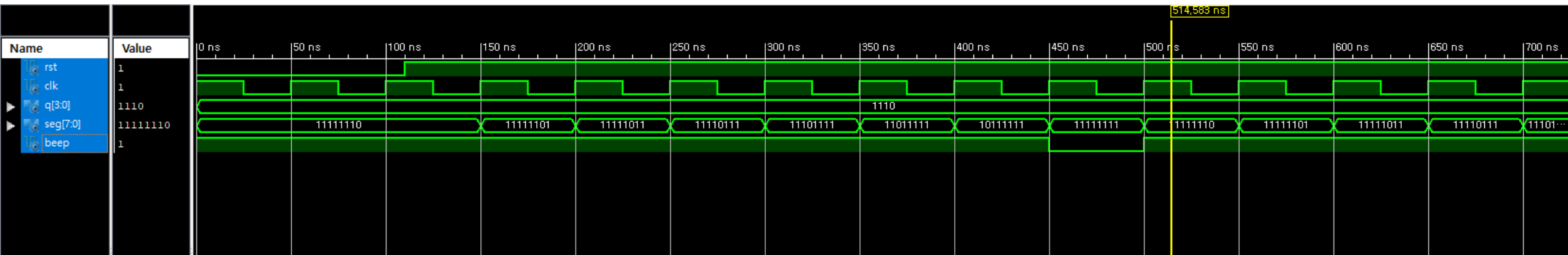
비동기식 reset + 8진 BCD+ Beep 동작테이블

비동기식 Reset	Clock	beep	8진	INPUT-BCD		
EV3	CLK	Beep	내부 Count	C	B	A
J12	C9	E11		0	0	0
1	↑	1	0	0	0	0
1		1	0	0	0	0
1		1	1	0	0	1
1		1	2	0	1	0
1		1	3	0	1	1
1		1	4	1	0	0
1		1	5	1	0	1
1		1	6	1	1	0
1		1	7	1	1	1
0		0	0	0	0	0

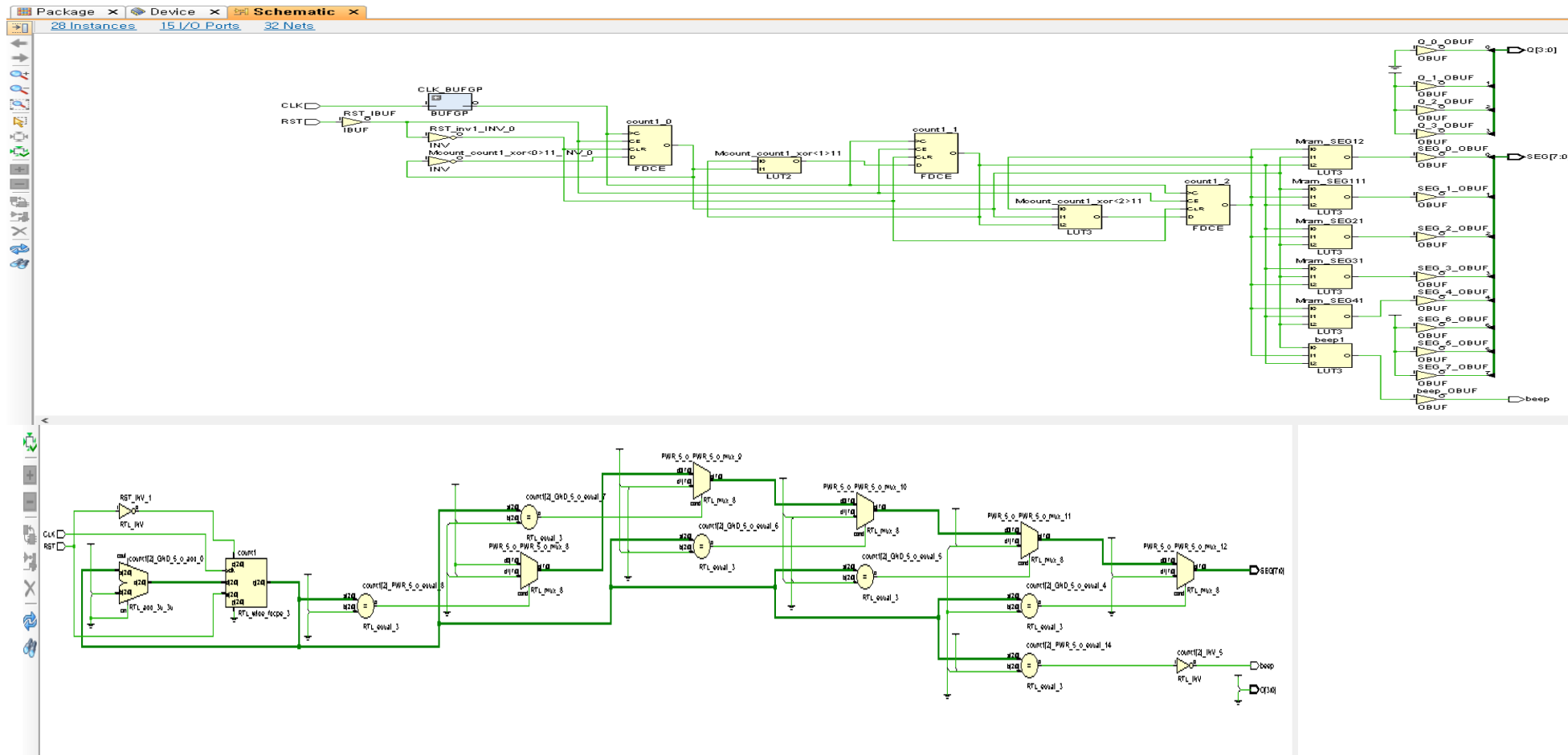
비동기식 reset + 8진 segment counter + Beep 동작테이블

16진 수	Common Anode 7-세그먼트LED 코드									Common Anode 7-세그먼트			
	dp	g	f	e	d	c	b	a	Beep	TR3	TR2	TR1	TR0
0	1	1	1	1	1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	0	1	1				
2	1	1	1	1	1	0	1	1	1				
3	1	1	1	1	0	1	1	1	1				
4	1	1	1	0	1	1	1	1	1				
5	1	1	0	1	1	1	1	1	1				
6	1	0	1	1	1	1	1	1	1				
7	1	1	1	1	1	1	1	1	0				

비동기식 reset + 8진 segment counter + Beep 시뮬레이션화면



비동기식 reset + 8진 segment counter + Beep 회로화면



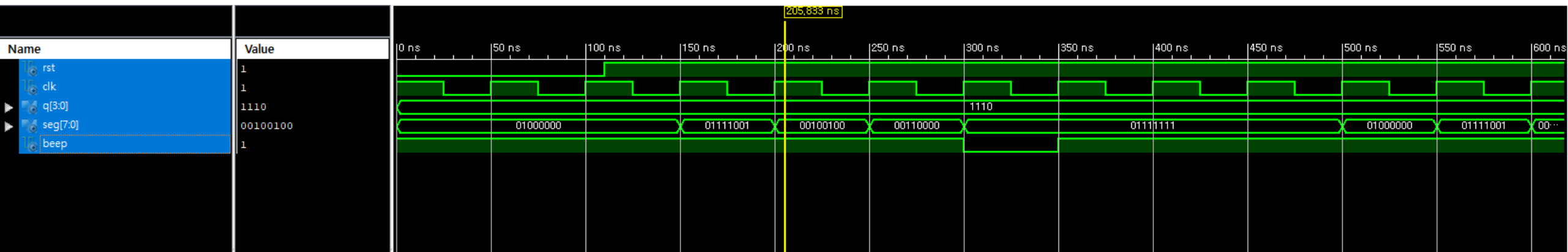
비동기식 reset + 5진 BCD + Beep 동작테이블

비동기식 Reset	Clock	beep	5진	INPUT-BCD		
EV3	CLK	Beep	내부 Count	C	B	A
J12	C9	E11		0	0	0
1	↑	1	0	0	0	0
1		1	0	0	0	1
1		1	1	0	1	0
1		1	2	0	1	1
1		1	3	1	0	0
1		1	4	1	0	1
0		0	0	0	0	0

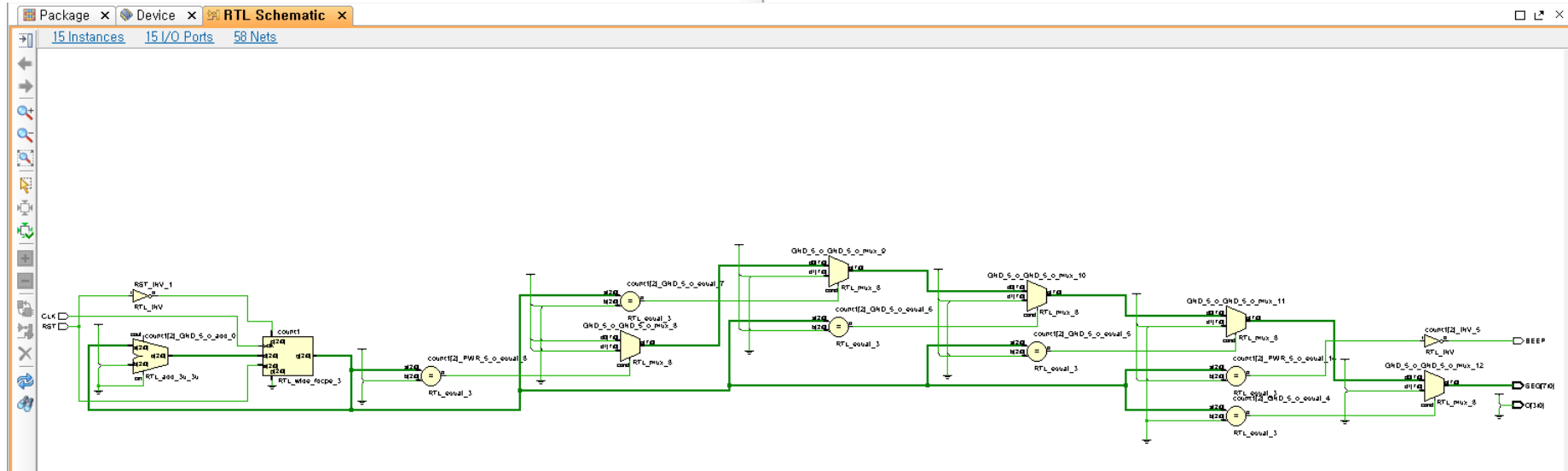
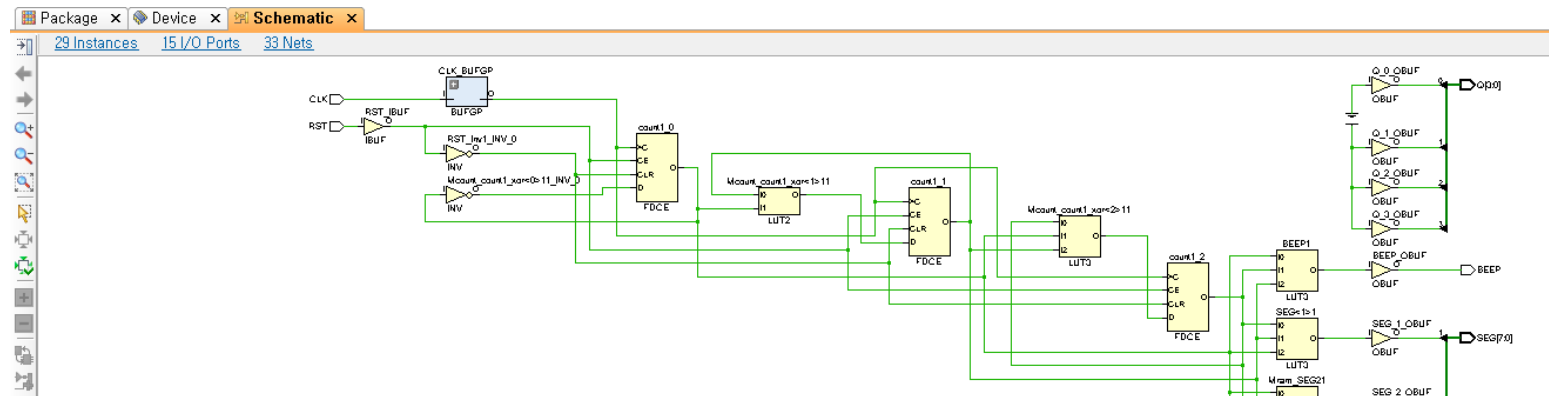
비동기식 reset + 5진 segment counter + Beep 동작테이블

16진 수	Common Anode 7-세그먼트LED 코드									Common Anode 7-세그먼트			
	dp	g	f	e	d	c	b	a	Beep	TR3	TR2	TR1	TR0
0	1	1	1	1	1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	0	1	1				
2	1	1	1	1	1	0	1	1	1				
3	1	1	1	1	0	1	1	1	1				
4	1	1	1	0	1	1	1	1	1				

비동기식 reset + 5진 segment counter + Beep 시뮬레이션화면



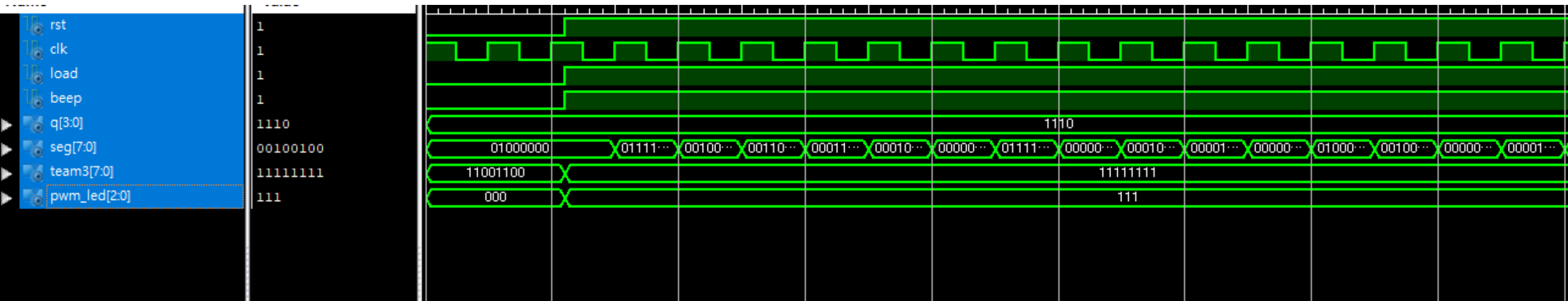
비동기식 reset + 5진 segment counter + Beep 회로화면



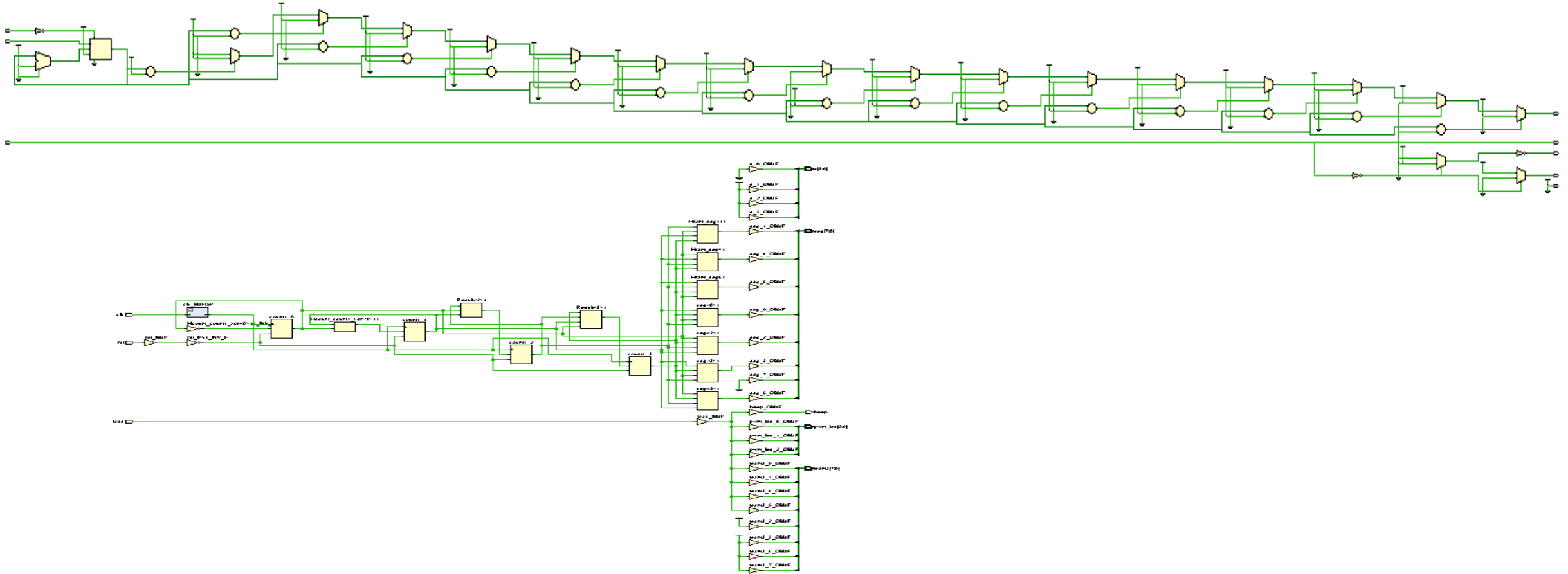
CLOCK	RESET	LOAD	COUNT	DP	G	F	E	D	C	B	A	TR3	TR2	TR1	TR0
CLK	RST	LOAD	COUNT1	A11	A9	E6	D6	C8	C6	B8	A8	D11	C13	B14	A13
C9	J12(EV3)	J14(EV2)	내부 신호									1	1	1	0
On	1	1	0	0	1	0	0	0	0	0	0				
			1	0	1	1	1	1	0	0	1				
			2	0	0	1	0	0	1	0	0				
			3	0	0	1	1	0	0	0	0				
			4	0	0	0	1	1	0	0	1				
			5	0	0	0	1	0	0	1	0				
			6	0	0	0	0	0	0	1	0				
			7	0	1	1	1	1	0	0	0				
			8	0	0	0	0	0	0	0	0				
			9	0	0	0	1	0	0	1	1				
			a	0	0	0	0	1	1	1	0				

		D	G	F	E	D	C	B	A
		P							
a	1	0	0	0	0	1	0	0	0
b		0	0	0	0	0	0	1	1
c		0	0	1	0	0	1	1	1
d		0	0	1	0	0	0	0	1
e		0	0	0	0	0	1	1	0
f		0	0	0	0	1	1	1	0

실습1 + 과제LOAD 기능추가 시뮬레이션 화면



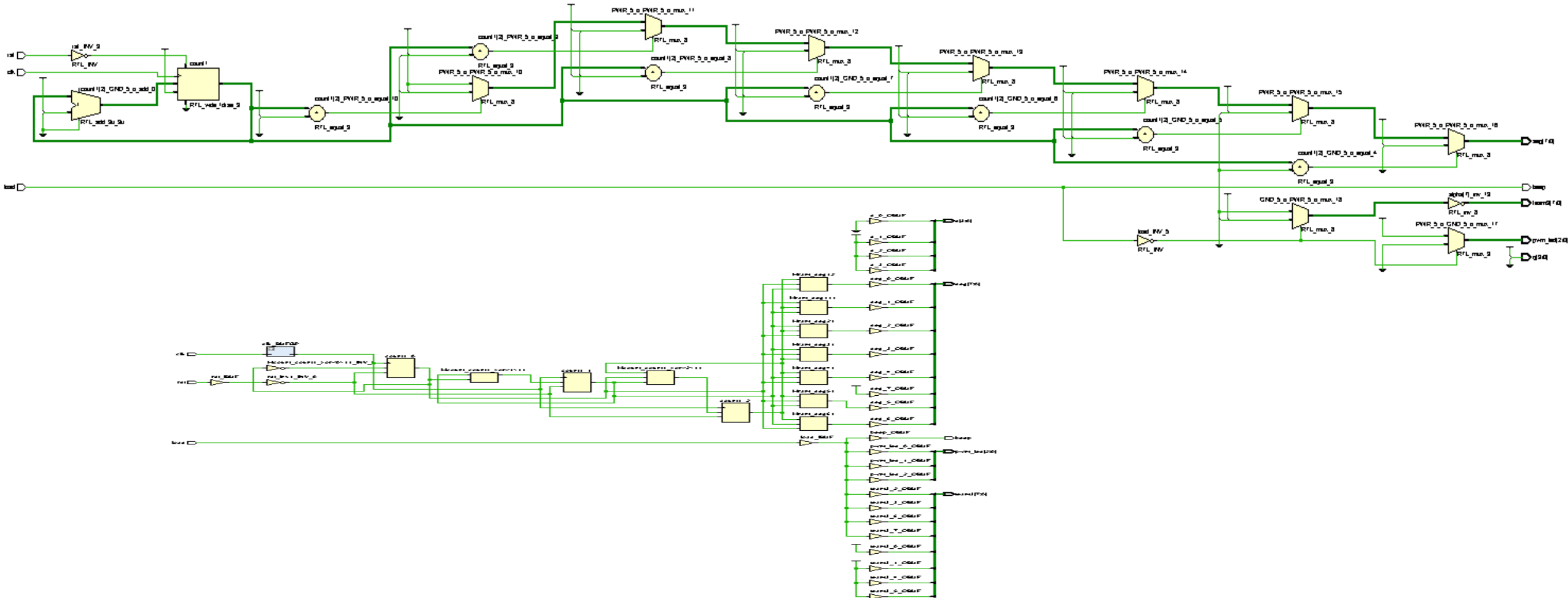
실습1 + 과제LOAD 기능추가 회로화면



실습2 + 과제LOAD 기능추가 동작테이블

CLOCK	RESET	LOAD	COUNT	DP	G	F	E	D	C	B	A	TR3	TR2	TR1	TR0	load	y5	y4	y1	y0
CLK	RST	LOAD	COUNT1	A11	A9	E6	D6	C8	C6	B8	A8	D11	C13	B14	A13	0	0	0	0	0
C9	J12(EV3)	J14(EV2)	내부신호									1	1	1	0					
On	1	1	0	1	1	1	1	1	1	1	0									
			1	1	1	1	1	1	1	0	1									
			2	1	1	1	1	1	0	1	1									
			3	1	1	1	1	0	1	1	1									
			4	1	1	1	0	1	1	1	1									
			5	1	1	0	1	1	1	1	1									
			6	1	0	1	1	1	1	1	1									
			7	0	1	1	1	1	1	1	1									

실습2 + 과제LOAD 기능추가 시뮬레이션 화면



실습2 + 과제LOAD 기능추가 회로화면

